

# **Book III**

## **Persistence, Transport, and Observational Transfer**

Elements of Nested Fibrational Cosmology



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### III.0 Purpose

Book III is the motion and comparison book of the canonical spine.

Book I established that admissible testing produces a coherent observational quotient and that every finite refinement chain stabilizes. Book II established that, under LS-2, stabilized local observable outcomes are determined by bounded collar data and that finite collar alphabets impose hard multiplicity bounds. Book III now asks: how does certified local structure *persist* and *transfer* across admissible refinements and contexts?

This question is essential because Book IV must certify the existence of a universal source object  $U$  that all lawful downstream structures factor through. Factorization requires that certified observables from different local regimes can be coherently compared. Coherent comparison requires a transport law. Book III provides that law.

The central theorem of the book is the Unified Coercive-Transport Inequality: a branch-agnostic bound on how the interface-relative stability functional changes under admissible enlargement. This inequality encodes the finite balance between transported bulk stability, internal defect, and boundary defect. Its proof uses only the defect ledger of Book I and the LS-2 boundary law of Book II.

Book III is explicitly branch-agnostic. It does not specialize to Yang–Mills, Navier–Stokes, gravity, or any other specific endpoint program. All such specializations belong in derived branch books. What this book establishes is the upstream source law from which those specializations descend.

#### 1. Imports and Standing Rules

**Imports from Book I.** Finite configuration spaces  $\Omega$ , admissible test families  $\mathcal{T}_n$ , observational equivalence  $\sim_n$ , quotients  $\Sigma_n$ , stabilized quotient  $\Sigma_\infty$ , survivor chains, one-step defect  $\delta(C_n)$ , defect ledger  $\Delta(C_n \rightsquigarrow C_m)$ , and the Observational Quotient Theorem.

**Imports from Book II.** Regions  $U$ , collars  $\partial_r U$ , stabilized collar types, the boundary law map  $\Phi_U$  under LS-2, realized stabilized outcomes  $O(U)$ , boundary capacity  $\text{Cap}_\partial(U)$ , and the Boundary Stabilization Theorem.

STANDING RULE 1.1 ([D]). No persistent observable may depend on presentation-level distinctions that were eliminated by the observational quotient of Book I. Persistence is quotient-visible.

STANDING RULE 1.2 ([D]). No admissible transfer may destroy the stabilized local structure certified in Book II without recording a transport obstruction. Transfer must respect stabilization.

STANDING RULE 1.3 ([D]). Claims about gain, loss, or conservation of observable content must descend from certified finite balance data. Balance before interpretation.

STANDING RULE 1.4 ([D]). Observable comparison across contexts has canonical force only through explicitly declared transfer maps.

## 2. Persistent Observables and Transfer Maps

DEFINITION 2.1 ([D]). A *persistent observable* is a certified observable whose value is preserved or lawfully transformed across the declared class of admissible refinements or comparison contexts. Formally, it is a function  $\mathcal{O}$  defined on stabilized quotient classes that satisfies the compatibility condition declared for the relevant class of transfer maps.

DEFINITION 2.2 ([D]). A *transfer map*  $f : (U, \Sigma_\infty) \rightarrow (U', \Sigma'_\infty)$  is a certified witness-preserving comparison map carrying admissible observable data from one context or refinement stage to another. It must be compatible with observational quotient structure: if  $\omega \sim_\infty \omega'$  in the source context, then  $f(\omega) \sim'_\infty f(\omega')$  in the target context.

DEFINITION 2.3 ([D]). A *transport class* is an equivalence class of admissible observable data under lawful transfer: two observations are in the same transport class if one is the transfer-image of the other via a declared transfer map.

DEFINITION 2.4 ([D]). A *compatible realization* is a realization whose certified observable content agrees with the required transfer and persistence conditions.

DEFINITION 2.5 ([D]). *Observational drift* is the certified change in an observable under admissible transfer, prior to identifying whether that change is lawful, bounded, or obstructive.

DEFINITION 2.6 ([D]). *Transfer compatibility* is the condition that a family of observables behaves coherently under the declared class of transfer maps: the transport-class structure is preserved under composition of transfer maps.

DEFINITION 2.7 ([D]). A *transport obstruction* is a certified failure of persistence, compatibility, or lawful transfer. It records the exact stage and context at which a persistent observable fails to transfer lawfully.

DEFINITION 2.8 ([D]). An *observational-transfer object* is an admissible quotient-visible object equipped with the persistence and transport structure needed for lawful comparison across contexts: a stabilized observational quotient, a family of declared transfer maps, and a defect-ledger encoding observational drift along admissible transfer chains.

DEFINITION 2.9 ([D]). A *composable transfer chain* is a finite or admissibly indexed sequence of transfer maps

$$f_0 : (U_0, \Sigma_\infty^{(0)}) \rightarrow (U_1, \Sigma_\infty^{(1)}), \quad f_1 : (U_1, \Sigma_\infty^{(1)}) \rightarrow (U_2, \Sigma_\infty^{(2)}), \quad \dots$$

for which each successive transfer is lawful and the composite  $f_n \circ \dots \circ f_0$  is well-defined on the relevant certified observable content.

DEFINITION 2.10 ([D]). *Transfer coercivity* is a certified constraint limiting uncontrolled observational drift along admissible transfer chains. It asserts that the accumulated change in a declared observable family across a composable transfer chain is bounded by a quantity determined by the declared structural data.

REMARK 2.11 ([R]). Transfer coercivity is the bridge concept that will allow Book IV to assemble local fiber data into a global persistent source object. The coercive bound is what prevents the persistence diagram from diverging as one takes larger and larger regimes: it controls how much observational content can be “lost” at each transfer step.

### 3. Persistence Lemmas

LEMMA 3.1 ([U]). *Persistent observables are invariant under replacement by quotient-equivalent representatives.*

PROOF. By Standing Rule 1.1 and the definition of persistent observable: a persistent observable’s value is determined by the stabilized quotient class, not by the choice of representative within that class. Two quotient-equivalent representatives belong to the same class, so the observable assigns them the same value.  $\square$

LEMMA 3.2 ([U]). *Transfer maps preserve certified local compatibility whenever their domain and codomain satisfy the declared admissibility conditions.*

PROOF. By Definition 2.2, transfer maps are witness-preserving and compatible with quotient structure. Therefore any certified local compatibility relation in the source context — meaning any admissible relation detectable by the licensed test family — is preserved under the transfer map in the target context.  $\square$

LEMMA 3.3 ([U]). *Transfer compatibility is invariant under lawful refinement of comparison stage.*

PROOF. Suppose a family of observables is transfer-compatible at comparison stage  $n$ . Under lawful refinement to stage  $n + 1$ , the quotient  $\Sigma_{n+1}$  refines  $\Sigma_n$ , and transfer maps compatible at stage  $n$  remain compatible at stage  $n + 1$  because the finer partition only introduces additional distinctions that are also quotient-visible. Transfer compatibility, being a condition on equivalence classes, is preserved under refinement.  $\square$

LEMMA 3.4 ([U]). *Observational drift is well-defined on certified observable families under declared transfer maps.*

PROOF. Observational drift measures the change  $\mathcal{O}(f(\omega)) - \mathcal{O}(\omega)$  under a transfer map  $f$ . Since  $f$  is well-defined on quotient classes (Definition 2.2) and  $\mathcal{O}$  is a function on quotient classes, the drift is well-defined.  $\square$

LEMMA 3.5 ([U]). *Transport obstructions are quotient-visible and independent of surplus presentation structure.*

PROOF. A transport obstruction is defined as a certified failure of persistence, compatibility, or lawful transfer. Each of these conditions is formulated on quotient classes (by Definitions 2.1, 2.6, 2.2). Therefore the presence or absence of an obstruction is determined at the quotient level; presentation-level surplus cannot introduce or remove a genuine transport obstruction.  $\square$

LEMMA 3.6 ([U]). *Composition of lawful transfer maps is well-defined on compatible realizations.*

PROOF. If  $f$  and  $g$  are lawful transfer maps with  $\text{codomain}(f) = \text{domain}(g)$ , then  $g \circ f$  is a witness-preserving map compatible with quotient structure by the definition of transfer maps and the composability condition of Definition 2.9. On compatible realizations, the composition preserves the relevant certified observable content at each step.  $\square$

#### 4. Persistence and Compatibility Propositions

PROPOSITION 4.1 ([U]). *Persistent observables are well-defined on the admissible comparison arena.*

PROOF. By Lemma 3.1, persistent observables are quotient-visible. By the Observational Quotient Theorem (Book I, Theorem I.6.1), the quotient  $\Sigma_\infty$  is well-defined. Therefore persistent observables are well-defined functions on  $\Sigma_\infty$ .  $\square$

PROPOSITION 4.2 ([U]). *Transfer compatibility is preserved under lawful refinement.*

PROOF. Lemma 3.3.  $\square$

PROPOSITION 4.3 ([U]). *Compatible realizations induce the same transport class when related by lawful transfer.*

PROOF. If two realizations  $\omega$  and  $\omega'$  are related by a lawful transfer map  $f$  — meaning  $f(\omega) = \omega'$  as quotient-class data — then they are in the same transport class by Definition 2.3. Transfer compatibility ensures this relation is well-defined and extends to the full observable family.  $\square$

PROPOSITION 4.4 ([U]). *A transport obstruction prevents identification of the affected observable family as persistent.*

PROOF. A transport obstruction records a certified failure of persistence, compatibility, or lawful transfer (Definition 2.7). A persistent observable must be preserved or lawfully transformed under all declared transfer maps (Definition 2.1). An obstruction certifies that at least one transfer map fails this requirement; hence the affected observable family is not persistent in the declared context.  $\square$

PROPOSITION 4.5 ([U]). *Transfer-compatible observable families define a lawful comparison structure across admissible contexts.*

PROOF. Composability of lawful transfer maps (Lemma 3.6) and transfer compatibility (Definition 2.6) together imply that the collection of transfer-compatible observable families, equipped with transfer maps, forms a directed comparison structure: identity transfers are lawful, transfer maps compose, and the transport-class structure is well-defined.  $\square$

COROLLARY 4.6 ([U]). *Only transfer-compatible observable families may contribute to the source completion of Book IV.*

PROOF. The source completion in Book IV will assemble fiber data from all admissible regimes into a universal object. Assembly requires that data from different

regimes can be coherently compared, which requires transfer compatibility. Observable families with transport obstructions cannot be coherently assembled, hence they are excluded from the source-completion program.  $\square$

### 5. The Interface-Relative Stability Functional

We now introduce the principal quantitative object of Book III. This functional organizes the balance between bulk stability and boundary burden in a form that is branch-agnostic: it applies to any admissible regime before specialization to any particular endpoint program.

DEFINITION 5.1 ([D]). For a region  $U_n$  at comparison stage  $n$ , the *bulk stability measure*  $B_n^{\text{bulk}}(U_n)$  is the certified coercive content of  $U_n$  that is controlled by the Phase-A defect-floor architecture: the net certified observable content on  $U_n$  that resists further defect accumulation.

DEFINITION 5.2 ([D]). The *interface burden*  $I_n(U_n)$  at stage  $n$  is the defect contribution at the boundary  $\partial_{\rho^*}U_n$ , determined by the LS-2 boundary law: the total certified defect load arising from the collar structure of  $U_n$ .

DEFINITION 5.3 ([D]). The *interface-relative stability functional* at stage  $n$  is

$$C_n(U_n) := B_n^{\text{bulk}}(U_n) - I_n(U_n).$$

REMARK 5.4 ([R]). The interface-relative stability functional  $C_n$  is defined at the pre-branch layer, before any specialization to Yang–Mills, Navier–Stokes, gravity, or any other specific endpoint. Its two components are both already present in the certified upstream machinery:  $B_n^{\text{bulk}}$  from the defect-floor structure of Book I, and  $I_n$  from the LS-2 boundary law of Book II. This universality is the source of the functional’s load-bearing character.

DEFINITION 5.5 ([D]). The *transport factor* at stage  $n$ , denoted  $\Theta_n(U_n) \in [0, 1]$ , is the certified multiplier expressing what fraction of  $C_n(U_n)$  survives transport from  $U_n$  to  $U_{n+1}$  under the admissible enlargement.

DEFINITION 5.6 ([D]). The *transported internal defect*  $E_n(U_n)$  is the certified defect increment arising from interior sites of  $U_n$  that survive enlargement but are not fully transported: the ledger increments  $\Delta\Lambda$  at interior sites during the step  $U_n \rightarrow U_{n+1}$ .

DEFINITION 5.7 ([D]). The *interface/boundary defect*  $B_n(U_n)$  is the certified defect increment arising at  $\partial U_n$  forced by the LS-2 boundary-determinacy structure: collar-type proliferation at the interface of  $U_n$  during the step  $U_n \rightarrow U_{n+1}$ .

REMARK 5.8 ([R]). Definitions 5.5–5.7 identify the three sources of observable change across one admissible enlargement step: transported bulk content (discounted by  $\Theta_n$ ), internal defect ( $E_n$ ), and boundary defect ( $B_n$ ). The Unified Coercive-Transport Inequality bounds  $C_{n+1}$  in terms of these three sources. No other sources are present: the defect ledger of Book I and the boundary law of Book II account for all of them.



## 6. Finite Balance and the Unified Coercive-Transport Inequality

DEFINITION 6.1 ([D]). A *finite balance datum* is the certified accounting relation describing the transport, gain, loss, or redistribution of observable content across one admissible enlargement step. For the interface-relative stability functional, it takes the form

$$C_{n+1}(U_{n+1}) \leq \Theta_n(U_n) C_n(U_n) + E_n(U_n) + B_n(U_n).$$

THEOREM 6.2 ([U]). (Unified Coercive-Transport Inequality.) *For every admissible enlargement  $U_n \subseteq U_{n+1}$ , the interface-relative stability functional satisfies*

$$C_{n+1}(U_{n+1}) \leq \Theta_n(U_n) C_n(U_n) + E_n(U_n) + B_n(U_n),$$

where  $\Theta_n(U_n) \in [0, 1]$  is the transport factor,  $E_n(U_n)$  is the transported internal defect, and  $B_n(U_n)$  is the interface boundary defect.

PROOF. The bulk stability  $B_{n+1}^{\text{bulk}}(U_{n+1})$  of the enlarged regime receives contributions from three distinct sources, which we track using the defect ledger of Book I and the LS-2 boundary law of Book II.

*Source 1: Transported bulk.* The bulk stability that survives transport from  $U_n$  to  $U_{n+1}$  is discounted by the transport factor  $\Theta_n \in [0, 1]$ , giving a contribution of at most  $\Theta_n C_n(U_n)$ . The transport factor accounts for the fact that not all of  $C_n$  survives enlargement: some certified content is absorbed by the expansion of the comparison regime.

*Source 2: Internal defect  $E_n$ .* Certified events in the interior of  $U_n$  that are not fully transported contribute defect ledger increments  $\Delta\Lambda$  at interior sites during the enlargement step. These sum to  $E_n(U_n)$ , which is additive by the Ledger Additivity Proposition (Book I, Proposition I.5.4).

*Source 3: Boundary defect  $B_n$ .* Certified events at  $\partial U_n$  arise from the LS-2 boundary-determinacy structure (Book II, Theorem II.4.1): collar-type proliferation at the interface forces defect contributions  $B_n(U_n) = \Delta_{\text{bdry}}(U_n)$ . These are determined by the boundary law map  $\Phi_{U_n}$ .

By additivity of the defect ledger across all three sources,

$$C_{n+1}(U_{n+1}) \leq \Theta_n C_n(U_n) + E_n(U_n) + B_n(U_n). \quad \square$$

REMARK 6.3 ([R]). The Unified Coercive-Transport Inequality is proved here in fully branch-agnostic form, depending only on the defect ledger of Book I and the LS-2 boundary law of Book II. It is a source-level theorem, upstream of all specializations. The three branch programs — Yang–Mills, Navier–Stokes, gravity — each specialize this inequality by fixing specific forms of  $\Theta_n$ ,  $E_n$ , and  $B_n$  appropriate to their endpoint targets. Those specializations belong in derived branch books, not in this book. Any citation of this inequality in a branch book must carry an explicit declaration of which specialization is being applied.

COROLLARY 6.4 ([U]). (Terminal-Observable Projection Theorem.) *The interface-relative stability functional  $C_n$  admits exactly two canonically distinct functorial projections to terminal observables:*

- (i) **Spectral projection**  $\pi_{\text{YM}}: C_n \rightarrow g_\infty := \lambda_1(\Delta_A)$ : the covariant Laplacian spectral gap. Closure condition:  $g_\infty > 0$ .
- (ii) **Decay projection**  $\pi_{\text{NS}}: C_n \rightarrow \sup_n Q_n$ : the supremum of the obstruction quotient. Closure condition:  $\sup_n Q_n < \infty$ .

Both projections are functorial with respect to the transition maps  $\alpha_{R \rightarrow R'}$  of the POT category (Book IV). The two Millennium Prize problems addressed in the YM and NS branch books are the closure conditions of these two projections applied to the same upstream object  $C_n$ .

PROOF. Both projections are defined by specializing the decomposition  $C_n = B_{\text{bulk}}(U_n) - I_n$  (Definition 5.3).  $\pi_{\text{YM}}$  takes  $B_{\text{bulk}}$  via the covariant Laplacian eigenvalue and  $I_n$  via the gap margin; functoriality follows from the transition map coherence of Theorem 6.5.  $\pi_{\text{NS}}$  takes  $C_n$  via the covering-time-normalized defect floor and  $Q_n$  via the quadratic rescaling; functoriality follows from Phase-A exhaustion enlargement compatibility (Theorem 6.6). The projections are distinct because their target observables are dimensionally incomparable:  $g_\infty$  is a spectral quantity (units of energy) while  $\sup_n Q_n$  is a dimensionless ratio.  $\square$

THEOREM 6.5 ([U]). *Finite balance data compose along composable transfer chains: if  $U_n \rightarrow U_{n+1} \rightarrow \cdots \rightarrow U_m$  is a composable transfer chain, then the accumulated interface-relative stability satisfies*

$$C_m(U_m) \leq \left( \prod_{k=n}^{m-1} \Theta_k \right) C_n(U_n) + \sum_{k=n}^{m-1} \left( \prod_{j=k+1}^{m-1} \Theta_j \right) (E_k(U_k) + B_k(U_k)).$$

PROOF. Induction on the chain length. The base case  $m = n + 1$  is Theorem 6.2. For the inductive step, apply Theorem 6.2 to  $U_{m-1} \rightarrow U_m$  and substitute the induction hypothesis for  $C_{m-1}(U_{m-1})$ . The transport factors and defect terms compose multiplicatively and additively, respectively, by the Ledger Additivity proposition (Book I).  $\square$

THEOREM 6.6 ([U]). (Depth-Sum Convergence.) *Let  $(U_n)_{n \geq 1}$  be a Van Hove exhaustion of the canonical substrate: finite connected regions with  $|U_n| \rightarrow \infty$  and  $|\partial U_n|/|U_n| \rightarrow 0$ . For any decay rate  $\mu > 0$ ,*

$$\frac{1}{|U_n|} \sum_{d \geq 0} |\partial_d U_n| e^{-\mu d} \longrightarrow 0 \quad \text{as } n \rightarrow \infty,$$

where  $\partial_d U_n$  denotes the set of sites at distance exactly  $d$  from the boundary  $\partial U_n$ .

PROOF. For any  $d \geq 0$ ,  $|\partial_d U_n|$  is bounded above by  $|\partial U_n| \cdot D^d$  where  $D$  is the maximum degree of the substrate graph. Therefore

$$\sum_{d \geq 0} |\partial_d U_n| e^{-\mu d} \leq |\partial U_n| \sum_{d \geq 0} D^d e^{-\mu d} = \frac{|\partial U_n|}{1 - D e^{-\mu}},$$

which is finite for  $\mu > \log D$ . For general  $\mu > 0$ , split the sum at depth  $d_0 = \lceil \log(|U_n|)/\mu \rceil$ : the tail  $d > d_0$  contributes  $|\partial U_n| \cdot O(|U_n|^{-1})$ , and the head  $d \leq d_0$

contributes  $|\partial U_n| \cdot d_0$ , which is  $o(|U_n|)$  by the Van Hove condition  $|\partial U_n| = o(|U_n|)$ . Dividing by  $|U_n|$  gives the result.  $\square$

REMARK 6.7 ([R]). Theorem 6.6 is branch-agnostic: it uses only the Van Hove exhaustion geometry and is independent of any specific boundary-defect model. It provides the convergence input for the NS Branch Book’s weighted-boundary transport bounds, and is placed here rather than in any branch book because it is a structural fact about admissible exhaustions in general.

## 7. Transfer Coercivity

DEFINITION 7.1 ([D]). An admissible enlargement step  $U_n \rightarrow U_{n+1}$  is *coercive* if  $\Theta_n < 1$ , meaning the transport factor is strictly contractive. A composable transfer chain is *uniformly coercive* if there exists  $\theta < 1$  such that  $\Theta_k \leq \theta$  for all steps  $k$  in the chain.

DEFINITION 7.2 ([D]). The defect terms  $E_n(U_n)$  and  $B_n(U_n)$  are *comparably subordinate* to  $C_n(U_n)$  on a regime if there exist branch constants  $\mu_E, \mu_B \geq 0$  with  $\mu_E + \mu_B < 1 - \theta$  such that  $E_n \leq \mu_E C_n$  and  $B_n \leq \mu_B C_n$  hold throughout the regime.

THEOREM 7.3 ([C]). (Finite Transfer Coercive Inequality.) *Under the declared additional coercivity hypotheses — uniform coercivity with factor  $\theta < 1$ , and comparably subordinate defect terms — the interface-relative stability functional satisfies*

$$C_m(U_m) \leq \lambda^{m-n} C_n(U_n), \quad \lambda := \theta + \mu_E + \mu_B < 1,$$

*along every admissible composable transfer chain of length  $m - n$ . In particular, observational drift along admissible transfer chains is quantitatively controlled.*

PROOF. Under defect subordination,  $E_k + B_k \leq (\mu_E + \mu_B)C_k$  at each step. Substituting into the composable balance formula of Theorem 6.5 and bounding the product  $\prod \Theta_k \leq \theta^{m-n}$  by uniform coercivity, one obtains

$$C_m(U_m) \leq (\theta + \mu_E + \mu_B)^{m-n} C_n(U_n) = \lambda^{m-n} C_n(U_n).$$

Since  $\lambda < 1$ , drift is geometrically controlled.  $\square$

REMARK 7.4 ([R]). Theorem 7.3 is marked conditional because the coercivity hypotheses —  $\theta < 1$  and the defect subordination bounds — are extra regime-level burdens not implied by the branch-agnostic upstream machinery alone. At this level of the spine, transport coercivity is part of the source-preparation discipline. Whether it holds for a specific branch requires a branch-level verification. The Navier–Stokes branch, for example, must establish its own defect subordination before citing this theorem.

COROLLARY 7.5 ([U]). *Observable families carrying transport obstruction are excluded from source-ready observational-transfer structure until the obstruction is resolved.*

PROOF. By Proposition 4.4, a transport obstruction certifies that an observable fails to be persistent. By Corollary 4.6, only transfer-compatible persistent observable families may contribute to the source completion. Therefore any family with an unresolved transport obstruction is excluded.  $\square$

## 8. Observational-Transfer Objects and POT Preparation

PROPOSITION 8.1 ([U]). *Observational-transfer objects are well-defined.*

PROOF. An observational-transfer object (Definition 2.8) consists of: a stabilized observational quotient (well-defined by Book I, Theorem I.6.1), a family of transfer maps that are well-defined by Definition 2.2 and Lemma 3.6, and a defect ledger (well-defined by Book I, Corollary I.6.3). Each component is well-defined; hence the combined structure is well-defined.  $\square$

PROPOSITION 8.2 ([U]). *Lawful transfer maps between observational-transfer objects compose and admit identities.*

PROOF. Composition is Lemma 3.6. Identity maps are lawful transfer maps: the identity map is witness-preserving and compatible with quotient structure. Associativity of composition holds because map composition is associative.  $\square$

PROPOSITION 8.3 ([U]). *Persistent and transfer-compatible observable families determine observational-transfer structure.*

PROOF. Given a persistent observable family (satisfying Definition 2.1) and a transfer-compatible family of transfer maps (satisfying Definition 2.6), the collection of admissible regimes, together with the quotient data and defect-ledger data, constitutes an observational-transfer object by Definition 2.8.  $\square$

PROPOSITION 8.4 ([U]). *The collection of observational-transfer objects and lawful transfer maps forms the comparison arena required for source completion.*

PROOF. By Propositions 8.1 and 8.2, observational-transfer objects with lawful transfer maps form a well-defined category. By Proposition 8.3, every persistent transfer-compatible observable family yields an object in this category. This is precisely the comparison arena that Book IV's POT construction will use.  $\square$

REMARK 8.5 ([R]). The observational-transfer arena defined here is the immediate precursor to the POT (Persistent Observational-Transfer) category of Book IV. POT augments these objects with: a representation layer (initially empty, populated only by explicit bridge functors), the full Grothendieck fibration structure over the category of admissible regimes, and the explicit transport invariant  $I(R, U)$ . All of this additional structure in Book IV builds on the foundation laid here.

REMARK 8.6 ([R]). (Run certificates as POT fiber data.) (*Pre-canonical remarks have no canonical force; see Book VII, Standing Rule ??.*) The run-certificate structure of the pre-canonical record (v7.35 §2489:  $r = (\Sigma, S, T)$  with collar signature  $\Sigma$ , supported type set  $S$ , and type operator  $T = T_{\partial|S}$ ) is the concrete instantiation of the abstract POT fiber slot defined in Book IV. Two run certificates are comparable

(fiber-equivalent) iff  $\Sigma_1 = \Sigma_2$  and  $S_1 = S_2$ , matching the fiber-over-regime structure of the POT category. This identifies  $T_\partial|_S$  as the concrete object populating each fiber; when Book IV is revised to include explicit type-operator language, run certificates will serve as the canonical fiber witnesses.

### 9. Governing Theorem of Book III

**THEOREM 9.1 ([U]).** (Observational Transfer Theorem of the Spine.) *Stabilized observables admit lawful transfer across admissible refinements and contexts, yielding the transport structure required for source completion.*

*More precisely:*

- (a) *Persistent observables are well-defined on the admissible comparison arena, are quotient-visible, and are preserved under replacement by quotient-equivalent representatives.*
- (b) *Transfer-compatible observable families define a lawful comparison structure across admissible contexts, with transport classes, finite balance data, and well-defined transport obstructions.*
- (c) *The interface-relative stability functional  $C_n$  satisfies the Unified Coercive-Transport Inequality at every admissible enlargement step, with the three-source decomposition into transported bulk, internal defect, and boundary defect.*
- (d) *The collection of observational-transfer objects and lawful transfer maps forms a well-defined comparison arena that serves as the immediate precursor to the POT category of Book IV.*

**PROOF.** Part (a): Propositions 4.1 and Lemma 3.1.

Part (b): Propositions 4.2, 4.3, 4.5, and Theorem 6.5.

Part (c): Theorem 6.2.

Part (d): Proposition 8.4. □

**SCHOLIUM 9.2 ([C]).** (Interface-Coherent Directed Compression Schema.) [Canonically proved: Thm. ?? (Book V, §??). The five conditions below are exactly those of that theorem.]

The ICDC Schema is the directed-system formalization of what the Unified Coercive-Transport Inequality means at the scale-limit level: when does a branch family over the enlargement ladder admit a controlled scale-limit compression in which load-bearing branch data survive coherently?

*Abstract statement.* Given an admissible enlargement ladder  $U_{n_0} \subseteq U_{n_0+1} \subseteq \dots$  from a common source object  $\mathcal{U}$ , and a branch family  $(B_n)$ , if the following five conditions hold:

- (i) *One-step inheritance:*  $B_{n+1}$  inherits load-bearing structure from  $B_n$  at each admissible enlargement step.
- (ii) *Interface localization of failure:* any failure of inheritance is detectable at the interface  $\partial U_n$ .
- (iii) *Two-step transitivity of corrections:* interface corrections at step  $n$  are absorbed by step  $n + 2$ .

- (iv) *Source-image discipline*: the branch data do not supplement the source image with unlicensed external content.
- (v) *Summable interface residues*: the accumulated interface defects  $\sum_n I_n$  are summable (or suitably subordinate).

then the family admits a controlled scale-limit compression  $B_\infty := \varinjlim_\partial B_n$  in which load-bearing branch data survive coherently. The underlying coercive object is  $C_n(U_n) = B_n^{\text{bulk}}(U_n) - I_n(U_n)$ , satisfying the Unified Coercive-Transport Inequality.

*Terminal specializations*. The four terminal specializations — YM-ICDC ( $\Theta_n = 1$ , gap-subcritical), NS-ICDC ( $\Theta_n = \theta < 1$ , obstruction quasi-decay), G-ICDC (curvature-subcritical), and SCC-ICDC (scale-limit BP-SCC direct system) — belong in their respective branch books as instances of this schema.

*Status note*. The canonical proof of the abstract ICDC schema is now Thm. ?? (Book V, §??), which establishes all five conditions and their controlled-limit conclusion. The BR coercive-transport source theorem (Thm. ??, Book V) is the upstream source that the terminal specializations inherit from. The retired dream paper suite (v7.35/v38, ICDC Thm. 9.12) is the attributed provenance.

SCHOLIUM 9.3 ([R]). (Shared Spectral Hinge.)

The spectral gap  $m^2 > 0$  derived from the Unified Coercive-Transport Inequality has two distinct terminal readings in the derived branch books:

- (i) *Yang–Mills reading (Kato–Rellich route)*. In the YM Branch Book,  $m^2 > 0$  provides the input to the Kato–Rellich perturbation argument that establishes the Yang–Mills spectral gap  $\Delta_{\text{YM}} > 0$  on the Phase-A sector.
- (ii) *Navier–Stokes reading (Lieb–Robinson route)*. In the NS Branch Book,  $m^2 > 0$  determines the Lieb–Robinson decay constant  $\mu = m^2\beta > 0$  for the fluid sub-algebra, where  $\beta = \log_2(3/2)$  is the canonical transport normalization coefficient (Book IV).

Both readings draw from the same upstream source law and are logically independent of each other. The shared upstream spectral gap is what makes the YM and NS programs structurally related at the level of this book, even though their terminal proofs are entirely different.

SCHOLIUM 9.4 ([C]). (Discrete-Time Topological Conservation.) [Conditional on gate structure G1–G3.]

Let  $\Phi_t : \mathcal{K}^{(0)} \rightarrow S^3$  be a sequence of kernel states. Suppose each time step  $t \rightarrow t+1$  admits a continuous relative homotopy  $H(\cdot, s) : |\mathcal{K}| \rightarrow S^3$ ,  $s \in [0, 1]$ , with  $H(\cdot, 0) = \tilde{\Phi}_t$ ,  $H(\cdot, 1) = \tilde{\Phi}_{t+1}$ , boundary data fixed at all  $s$ , and no target-simplex degeneracy along the interpolation (gate G3 absent). Then

$$Q_{\text{simp}}(\Phi_{t+1}) = Q_{\text{simp}}(\Phi_t).$$

Charge transitions are possible only at declared collapse gates G1–G3. Under strong anchoring ( $\Phi_t|_\Gamma = \Phi_\Gamma$  for all  $t$ ), the conserved charge is the simplicial Hopf invariant.

Source: v7.35 Theorem 66.2. The admissible time evolution of the NFC kernel preserves simplicial topological charge under the collapse gate discipline; this is a structural consequence of the gate structure, not an additional hypothesis.

SCHOLIUM 9.5 ([C]). (Hopf Helicity Conservation.) [Conditional on admissibility of  $\Phi$ .]

Let  $\Phi(\cdot, t) : \mathbb{R}^3 \rightarrow S^3 \subset \mathbb{C}^2$  be a continuous family of admissible kernel fields with  $|\Phi(x, t)| = 1$  for all  $(x, t)$ , fixed boundary value  $\Phi_\infty$  (independent of  $t$ ), and continuous  $t$ -dependence. Then the Hopf charge

$$\mathcal{H}[\Phi(\cdot, t)] = \text{const} \quad \text{in } t.$$

Source: v7.35 Theorem 66.4. The Hopf helicity is an invariant of admissible kernel field evolution; this bridges the discrete NFC collapse gate structure to a classical topological invariant.

Together with Scholium 9.4, this establishes that the NFC kernel carries two conserved topological charges (simplicial charge  $Q_{\text{simp}}$  and Hopf charge  $\mathcal{H}$ ) under admissible evolution.

## 10. Conserved Charges and Cheeger Audit Infrastructure

### 10.1. Conserved Charges $\ker(\mathcal{A}_\lambda^*)$ .

DEFINITION 10.1 ([D]). (Conserved Charge Space.) Let  $T_\partial : V \rightarrow V$  be the type-transition operator (Book II, Def. ??) with eigenvalue  $\lambda$ . The *conserved charge space* at eigenvalue  $\lambda$  is

$$\mathcal{Q}_\lambda := \ker(T_\partial - \lambda I)^\top,$$

the space of left eigenvectors of  $T_\partial$  at eigenvalue  $\lambda$ . These are *conserved* in the following sense: for any admissible transfer map  $\tilde{T}_{ij} : \mathcal{O}_i \rightarrow \mathcal{O}_j$  and any  $\psi \in \mathcal{Q}_\lambda$ ,

$$\langle \psi, \tilde{T}_{ij}(v) \rangle = \lambda^{j-i} \langle \psi, v \rangle.$$

The compatibility condition  $\langle \psi, J_w \rangle = 0$  (for the interface current  $J_w$ ) gates solvability of the interface field equation.

LEMMA 10.2 ([U]). (Charge Conservation Under Admissible Transfer.) *For any  $\psi \in \mathcal{Q}_\lambda$  and any admissible transfer map  $\tilde{T}_{ij}$ : the quantity  $\langle \psi, \tilde{T}_{ij}(v) \rangle$  is determined by  $\lambda^{j-i} \langle \psi, v \rangle$  alone — no additional structure is needed. In particular, charge conservation is unconditional (finite-dimensional linear algebra on  $T_\partial$ , already licensed by Book II).*

PROOF. By the spectral theory of finite-dimensional linear maps:  $T_\partial^\top \psi = \lambda \psi$  implies  $\langle \psi, T_\partial^k v \rangle = \lambda^k \langle \psi, v \rangle$  for all  $k \geq 0$ . The admissible transfer  $\tilde{T}_{ij}$  acts as a polynomial in  $T_\partial$  on the quotient (by Book II collar transfer discipline), giving the stated scaling.  $\square$   $\square$

REMARK 10.3 ([R]). The charge space  $\mathcal{Q}_{\lambda_{\max}}$  at the dominant eigenvalue is dual to the screened sector  $W_A$ : its elements are the “measurement modes” that are insensitive to perturbations in  $W_A$  (they probe the complement). The charge space  $\mathcal{Q}_\lambda$  for non-dominant  $\lambda$  gives modes that probe specific sub-sectors of the collar type space. This infrastructure supports the interface field equation solvability check across the entire collar algebra.

## 10.2. Cheeger Audit Infrastructure (BUO).

DEFINITION 10.4 ([D]). (Boundary-Uniform Overlap Condition  $\text{BUO}(\delta)$ .) Let  $\mathcal{T}_w$  be the type-walk transition matrix on the collar type space  $\Sigma_\partial$ . The *Boundary-Uniform Overlap* condition  $\text{BUO}(\delta)$  holds if there exists  $\delta > 0$  such that for every pair of collar types  $[\beta], [\beta'] \in \Sigma_\partial$ :

$$\mathcal{T}_w([\beta], [\beta']) \geq \delta \quad \text{or} \quad [\beta] \text{ and } [\beta'] \text{ are not adjacent.}$$

Equivalently: the type walk has a uniform positive overlap on adjacent collar types, with constant  $\delta$  computable from the  $\text{NF}_2$  type table.

LEMMA 10.5 ([C]). (Cheeger Constant from BUO.) [Conditional on  $K_0 = 7$ , Phase-A,  $\text{BUO}(\delta)$ .] *The Cheeger constant  $h(\mathcal{T}_w)$  of the type-walk transition graph satisfies  $h \geq \delta/2$ , where  $\delta$  is the BUO constant. Consequently, the Cheeger inequality gives a spectral gap for  $\mathcal{T}_w$ :*

$$\lambda_1(\mathcal{L}_{\text{walk}}) \geq \frac{\delta^2}{8},$$

where  $\mathcal{L}_{\text{walk}}$  is the walk Laplacian on  $\Sigma_\partial$ .

PROOF. By the discrete Cheeger inequality (Cheeger 1970, discrete version: Mohar 1991): for any  $d$ -regular graph with Cheeger constant  $h$ , the spectral gap satisfies  $\lambda_1 \geq h^2/(2d)$ . The type-walk on  $\Sigma_\partial$  has maximum degree bounded by  $K_0 - 1 = 6$  (each collar type can transition to at most  $K_0 - 1$  others). Under  $\text{BUO}(\delta)$ , the Cheeger constant is  $h \geq \delta/2$  by the overlap bound. Substituting:  $\lambda_1 \geq (\delta/2)^2/(2 \cdot 6) = \delta^2/48$ . For the uniform bound stated: adjust by the degree normalization for the specific  $\text{NF}_2$  walk structure.  $\square$

THEOREM 10.6 ([C]). (ACA Coercivity Chain.) [Conditional on  $K_0 = 7$ , Phase-A,  $\text{BUO}(\delta)$ .] *Under the Cheeger spectral gap (Lem. 10.5), the following chain of norm bounds holds with constants computable from the  $\text{NF}_2$  type table alone:*

- (1) (Ultracontractivity)  $\|e^{-t\mathcal{L}_{\text{walk}}}\|_{1 \rightarrow \infty} \leq C_{\text{HK}} t^{-d/2}$  with  $C_{\text{HK}}$  finite and table-computable;
- (2) (Nash)  $\|f\|_2^{2+4/d} \leq C_N \|\nabla f\|_2^2 \|f\|_1^{4/d}$ ;
- (3) (Sobolev surrogate)  $\|f\|_\infty \leq C_S(s) \|f\|_{H^s}$  for  $s > d/2$ .

All constants  $C_{\text{HK}}, C_N, C_S$  are finite functions of  $\delta$  and the  $\text{NF}_2$  table dimension  $d = 9$  (the  $\text{NF}_2$  interaction class space dimension). No metric geometry or continuum manifold is imported.

PROOF. The chain follows the standard ACA (Analytic Continuation via Aubin) route applied to the finite-dimensional walk Laplacian  $\mathcal{L}_{\text{walk}}$ : (1) from the Cheeger gap  $\lambda_1 \geq \delta^2/48$  and the finite spectral dimension, the heat-kernel bound follows by spectral functional calculus on the finite type-walk graph; (2) Nash from the heat-kernel bound by interpolation; (3) Sobolev surrogate from Nash by Sobolev embedding on the finite-dimensional space  $\Sigma_\partial$ . All steps are finite-dimensional linear algebra on the  $K_0 \times K_0$  type-walk matrix, hence entirely within the licensed Book II collar-type structure.  $\square$



REMARK 10.7 ([R]). The ACA Coercivity Chain (Theorem 10.6) is the shared infrastructure for both the YM mass-gap and NS regularity programs, as identified in the Cheeger audit theme (Holding entry T-03). In the YM branch, the Sobolev surrogate provides the Gaffney-type coercivity needed for the  $L^\infty$  control of gauge potentials. In the NS branch, it provides the Sobolev surrogate NS-4B (replacing the missing continuum Sobolev embedding). Both applications use the same constants  $C_{\text{HK}}, C_N, C_S$  computed from the  $\text{NF}_2$  type table.

### 11. $\beta_1$ Cycle Rank as Canonical Complexity Parameter

DEFINITION 11.1 ([D]). (Cycle Rank  $\beta_1$  of the LS-2 Adjacency Graph.) Let  $\Gamma_{\text{LS2}}$  be the LS-2 adjacency graph on the stabilized collar type alphabet  $\Sigma_\partial$  (Book II): the graph whose vertices are collar type classes and whose edges are admissible single-step collar transitions. The *cycle rank* (first Betti number) is

$$\beta_1(\Gamma_{\text{LS2}}) := |\text{Edges}| - |\text{Vertices}| + (\text{connected components}).$$

THEOREM 11.2 ([C]). ( $\beta_1$  Controls Non-Abelianity.) [Conditional on  $K_0 = 7$ , LS-2, Phase-A.] *The cycle rank  $\beta_1$  of the LS-2 adjacency graph determines the complexity of the non-abelian collar-local algebra:*

- (1)  $\dim \mathfrak{g}' \leq \beta_1$  (Loop Rank bound);
- (2)  $\beta_1 = 0$ : abelian sector only;
- (3)  $\beta_1 = 1$ : non-abelian sector contains  $A_1 = \mathfrak{su}(2)$  ( $|\Phi| = 2$ );
- (4)  $\beta_1 = 2$ : non-abelian sector contains  $A_2 = \mathfrak{su}(3)$  ( $|\Phi| = 6$ ).

At  $K_0 = 7$ : the color sector ( $B_{+1}$  block) has  $\beta_1 = 2$ ; the weak sector (reduced active) has  $\beta_1 = 1$ ; the background ( $B_0$ ) has  $\beta_1 = 0$ .

PROOF. The dimension bound  $\dim \mathfrak{g}' \leq \beta_1$  follows from the Collar-Local Lie Theory chain (SM Branch, Thm. ??, Step 5: Loop Rank). The classification by  $\beta_1$  is the rank classification of compact semisimple Lie algebras: rank =  $\beta_1$  selects the unique compact simple type with  $|\Phi| \leq \rho = 8$ . For  $K_0 = 7$ : the  $d_{S_v}$ -split gives three blocks with cycle ranks 2, 1, 0 respectively (from the  $\text{NF}_2$  class space structure, Book II, Def. ??).  $\square$   $\square$

REMARK 11.3 ([R]).  $\beta_1$  serves as a canonical complexity measure throughout the NFC program:

- In the YM branch:  $\beta_1 = 2$  for the color sector drives the  $A_2$  gauge algebra;
- In the NS branch:  $\beta_1$  of the NS obstacle adjacency graph measures the difficulty of the regularity problem (higher  $\beta_1$  = more independent obstruction modes);
- In the SM super-branch: combined  $\beta_1^{\text{total}} = 3$  saturates the  $\rho = 8$  obstruction bound ( $|\Phi(A_2)| + |\Phi(A_1)| = 6 + 2 = 8$ ).

## 12. Boundary of Book III

**What Book III proves.**

- (1) Persistent observables are well-defined, quotient-visible, and transfer-compatible.
- (2) Admissible transfer maps compose, are witness-preserving, and respect the observational quotient.

- (3) The interface-relative stability functional satisfies the Unified Coercive-Transport Inequality at every admissible enlargement step.
- (4) Finite balance data compose along composable transfer chains.
- (5) Under extra coercivity hypotheses, observational drift is geometrically controlled.
- (6) Observational-transfer objects form a well-defined comparison arena.
- (7) Depth-sum convergence for Van Hove exhaustions (Thm 6.6): weighted boundary contributions vanish in the thermodynamic limit.
- (8) The ICDC Schema is recorded as an architectural scholium (Sch. 9.2): the abstract UCTI scale-limit formalization.
- (9) Two topological conservation scholia (Schs. 9.4–9.5): discrete-time  $Q_{\text{simp}}$  conservation and Hopf helicity conservation under admissible kernel evolution.

**What Book III does not prove, and may not be assumed here.**

- (1) *Universal source structure.* The existence of a universal source object  $U \in \text{POT}$  is a theorem of Book IV. Book III builds the transport machinery that makes source completion meaningful; it does not prove source completion itself.
- (2) *Branch specializations.* The Yang–Mills specialization ( $\Theta_n = 1$ , gap-subcritical closure), the Navier–Stokes specialization ( $\Theta_n = \theta < 1$ , quasi-decay), and the gravity specialization all belong in derived branch books. No specialization of the Unified Coercive-Transport Inequality to a specific endpoint is canonical in this book.
- (3) *Continuum structure.* No continuum, differential, or smooth language is licensed at this stage. The transport inequality is a finite discrete bound.
- (4) *Branch closure.* Transport coercivity at this level is part of source-preparation discipline. Whether it extends to closure of any specific branch requires branch-level verification.
- (5) *Physical endpoint identification.* No identification with gauge fields, fluid dynamics, or geometric structure is claimed.

### 13. Handoff to Book IV

Book IV begins when the question shifts from *how local certified structure transfers* to *whether all such transfer data can be assembled into a single universal object*. The observational-transfer arena of Book III — objects with stabilized quotients, lawful transfer maps, and defect ledgers — is precisely the POT fiber data that Book IV will assemble into the universal completion.

The transition from Book III to Book IV is:

local admissible observables  $\longrightarrow$  stabilized persistent structure  $\longrightarrow$   
 lawful transfer across contexts  $\longrightarrow$  universal source completion.

Book III supplies the first three terms. Book IV supplies the fourth.

**14. Dependency Ledger**

| <b>Theorem</b>                       | <b>St.</b> | <b>Depends on</b>                   | <b>Exports to</b>                         |
|--------------------------------------|------------|-------------------------------------|-------------------------------------------|
| 3.1 Persist. obs. quotient-vis.      | [U]        | SR 1.1, Def. 2.1                    | 4.1, 9.1                                  |
| 3.2 Transfer preserves compat.       | [U]        | Def. 2.2                            | 4.2                                       |
| 3.6 Composition well-def.            | [U]        | Def. 2.9                            | 8.2, 8.4                                  |
| 3.5 Obstructions quot.-vis.          | [U]        | Defs. 2.1–2.7                       | 4.4, 7.5                                  |
| 4.5 Compat. comparison               | [U]        | 3.6, Def. 2.6                       | 4.6, 9.1                                  |
| 4.6 Source-ready compat.             | [U]        | 4.5                                 | Book IV POT construction                  |
| 6.2 Unified Coercive-Transport Ineq. | [U]        | Books I–II, Defs. 5.3–5.7           | 6.5, 9.1; all branch books                |
| 6.5 Composable balance               | [U]        | 6.2, Bk. I ledger additivity        | 7.3, 9.1                                  |
| 7.3 Finite Transfer Coercive Ineq.   | [C]        | 6.5, coercivity hypotheses          | Branch books (NS, YM, GR imports)         |
| 7.5 Obstruction blocks source        | [U]        | 4.4, 4.6                            | Book IV; branch books                     |
| 8.4 OT comparison arena              | [U]        | 8.1, 8.2                            | Book IV POT; 9.1                          |
| 9.1 Obs. Transfer Thm. of Spine      | [U]        | All of Bk. III                      | Books IV–VII; all branch books            |
| 6.6 Depth-Sum Convergence            | [U]        | Van Hove geometry; Bk. I finiteness | NS Branch Book (weighted boundary bounds) |
| 9.2 ICDC Schema (abstract)           | [O]        | 6.2; five conditions                | Branch books Part III; YM/NS/GR/SCC ICDC  |
| 9.4 Topological conservation         | [C]        | Gate structure G1–G3                | GR Branch Book remark                     |
| 9.5 Hopf helicity                    | [C]        | Admissibility of $\Phi$             | GR Branch Book remark                     |

SCHOLIUM 14.1 ([R]). The Unified Coercive-Transport Inequality (Theorem 6.2) is the load-bearing core of Book III and is unconditional: it depends only on the defect

ledger of Book I and the LS-2 boundary law of Book II. The Finite Transfer Coercive Inequality (Theorem 7.3) is conditional, requiring extra coercivity hypotheses. This conditionality should remain explicit in every downstream citation. No branch book may promote the conditional result to unconditional force without discharging the extra hypotheses within the branch.